

Harikrishna Pillai

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ABOUT ME

Student at Technische Universität Dresden, Germany, pursuing a Master's degree in Computational Modelling and Simulation. Passionate about Machine Learning and Deep Learning models, with a particular interest in generative models like **GANs**, **Diffusion** models and **Transformers**. Experimenting with **Neuroscience** and **Nature** inspired AI topics. Working on **Continual Learning** to create general and adaptive Neural Networks. Other interests include Game Development and 3D Graphics, with experience in working with **Unity**, **Godot**, and **Blender**.

EDUCATION

Technische Universität Dresden

Masters, Computational Modelling & Simulation

Dresden, Germany

2025 – present

- Expected Graduation Year : 2027

Amrita Vishwa Vidyapeetham

B. Tech, Computer Science and Artificial Intelligence

Kerala, India

2021 – 2025

- Expected Graduation Year : 2025
- CGPA : 8.87

MKVV International Vidyalaya

High Secondary

Mumbai, India

2019 - 2021

- Marks: 93%

EXPERIENCE

Research Intern @ Lossfunk, Bangalore, India

Jul 2025 – Present

- Designed and evaluated novel Continual Learning architectures using a scalable architecture called Tokenformer.
- Created and tested a few more hypotheses that aims to solve continual learning in different ways.

Research Intern @ CMC Lab, TU Dresden, Germany

Nov 2024 – Oct 2025

- Worked on studying neural activity of homeostatic RL agents to find interesting patterns with its foraging behaviour.
- Developing a weight perturbation algorithm that achieves convergence comparable to Gradient Descent.
- The results are promising and comparable to Gradient Descent.

ML Intern @ InSynk

Jan 2024 – June 2024

- Worked on creating custom E-commerce agents like product background generation, apparel background change, AI Ad Generation, Product Description tool etc. Used by E-commerce companies like flipkart, myntra, big bazar, reliance and many more. [product link](#)
- Worked on creating python backend and REST APIs. Also creating scripts that could generate fastapi code dynamically. [product link](#)

ML Intern @ Aveta.ai

August 2023 – Dec 2023

- Worked on making a special-purpose LLM for medical ICD coding.
- Used and explored techniques like RAG, Attention Manipulation and Prompt Engineering to optimize the LLM output.
- Created a tool that can analyze a PDF and intelligently separate all the patients data in different PDFs and save it.

Google Summer of Code 2022 @ INCF

June 2022 – Sep 2022

- Worked on mapping embryo microscopy images on a 3d ellipsoid generated using code. Created a Desktop application for its visualization.
- Used opencv library for image projection techniques and electron js for creating a desktop application to view and control the generated 3d object.

Mentor and Member @ amFOSS

Jan 2022 – present

- Actively contributing to open source software as a part of amFOSS, India's leading computer science club with over 60 passionate students.

RESEARCH

Homeostatic states shape neural and behavioral dynamics during foraging in embodied agents

- Worked to uncover the neural mechanisms underlying the observed behavior during foraging of Homeostatic RL agents.

CAN - Continuously Adapting Network

- We introduced CAN, a continual learning approach that combines Hebbian learning, selective gradient freezing, and adaptive network expansion. CAN leverages Hebbian plasticity to identify task-relevant neurons and freeze them to prevent forgetting. For each new task, CAN selectively reuses or grows the network by duplicating highly active neurons, enabling scalability and adaptability without catastrophic forgetting.

TALKS & POSTERS

Emergence of brain-like features in artificial neural networks trained based on homeostasis [\(@ BrainBody 2025\)](#)

- Talk to uncover the neural mechanisms underlying the observed behavior during foraging of Homeostatic RL agents.

Brain-like features in homeostatic deep reinforcement learning agents [\(@ Bernstein Conference 2025\)](#)

- Our model demonstrates the importance of homeostatic reward to study adaptive, survival-oriented behavior in ANNs. The observed coupling between nutrient levels, velocity, neural dynamics, and exploration patterns highlights how internal states modulate both cognition and neural dynamics.

PROJECTS

insuranceLLM

[Blog](#)

- The goal of this project is to make an NLP model that can understand a doctor's clinical report and output the relevant ICD (International Disease Classification) codes.
- These codes are then used by the insurance company to give the insurance money.
- Challenging because we only have the ICD guidelines as the tabular data PDF.
- Worked on both vector similarity and Attention manipulation as the potential solutions.

Skit Lite

[Source Code](#)

- This project aims to automate call centers using voice AI. The project uses OpenAI whisper and GPT models like chatGPT or LLaMA to make this possible.
- The GUI is entirely managed by Streamlit.

Doodle Diffusion

[Source Code](#)

- This project aims to build a diffusion based model that generates Google doodles from given prompts.
- The Reverse diffusion process is done using a UNet model which takes a noisy image, timestamp and text as inputs. The timestamp is embedded into the model using Sine Cosine Embedding and the text is encoded using a CLIP encoder and then added to the noisy image.

Cosmic Diffusion

[Source Code](#)

- This project uses a basic diffusion model to generate new stunning 64x64 nebula images that don't exist.

nano NeRF

[Source Code](#)

- It is a nano NeRF model that is trained to reconstruct 3D embryos of Axolotl using just few of its microscopic images.

DeepDino

[Source Code](#)

- Predicts the next move for the dinosaur in the Chrome Dino game using a CNN model.

TECHNICAL SKILLS

Languages	: Python, Java, JavaScript, Dart, Kotlin, MATLAB, C++, GDScript
Frameworks	: PyTorch, Django, Flutter, Docker
Libraries	: OpenCV, numpy, pandas, REST_API
Databases	: MongoDB, SQLite, Redis
OS	: Linux (Garuda, Ubuntu, Fedora, PoP), Windows
Dev Tools	: Visual Studio Code, Git, GitHub, Gitlab, blender, Godot

PROFESSIONAL REFERENCES

Dr. Jayraj Poroor

Kollam, India

Dean of School of Artificial Intelligence
Amrita Vishwa Vidyapeetham, Amritapuri
jayaraj.poroor@gmail.com

Abhijit Ramesh

Kollam, India

ML Engineer especially for Medical AI companies
<https://abhijitramesh.me/>
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EXTRACURRICULAR

Organized Workshop

- As a part of amFOSS, I organized an open source workshop for the first year students.
- During this 1 month long workshop, we taught them about Git, GitHub, Linux, Python, Web development and Android development.

NSEP (National Standard Examination in Physics)

- Cleared National Standard Examination in Physics and Qualified for Indian National Physics Olympiad

Captain of School Volleyball team

- Played as the captain of the school volleyball team and won second place in our district.